PARTCAST

AI-native demand planning for Aerospace Maintenance, Repair, and Overhaul (MRO) spare parts

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Problem Statement

01 MRO demand is forward-deterministic, not probabilistic.

A large share of future spare parts demand in aerospace Maintenance, Repair, and Overhaul (MRO) is knowable in advance. When an aircraft approaches a C-check, a scheduled heavy maintenance event, the required parts are already encoded in the maintenance program, the aircraft's flight-hours log, the airworthiness directives governing that airframe type, and the flight schedule driving its utilization rate. Compliance deadlines on those directives create hard replacement triggers on specific part numbers. Contract SLAs define which operators must be served and with what turnaround commitments.

The demand trigger is an engineering event with a known time horizon and a specific parts list attached. That makes it categorically different from the consumer or industrial demand that incumbent planning tools were built to sense.

02 Planners spend 60-70% of their review cycle on data reconciliation, not judgment.

Instead of evaluating demand signals, analysts spend the majority of every planning cycle manually pulling ERP exports, airline schedule feeds, FAA and EASA AD portals, supplier portals, and accumulated spreadsheets and reconciling them into a consensus forecast by hand. By the time that forecast reaches the supply chain team, it is already stale. A new AD was issued. A tail had an unscheduled removal. A supplier moved a delivery date. No one updated the plan. The process fails not because planners are bad at their jobs. It fails because the architecture of the tooling they use was designed around a consumption-history model of demand.

03 The consequence of getting it wrong is binary and catastrophic.

In most supply chains, a forecast miss degrades service levels, orders are delayed, customers are frustrated. In MRO, running out of a critical part grounds an aircraft. A grounded commercial aircraft costs an operator $150,000 or more per hour in lost revenue, crew costs, and passenger compensation. Many critical parts are single-source with 12-18 month lead times, no spot market, and in some cases export licensing requirements that add weeks to the procurement timeline. When you are out of the part, you are out. The cost of a forecast miss is not a premium freight charge, it is an aircraft on the ground for weeks.

Where Traditional Planning Tools Break Down

Traditional planning applications (such as Kinaxis and others) are powerful platforms built for industries where demand flows through point-of-sale systems and distributor orders. Their core architectural assumption that demand is a statistical property of time series does not fit well for aerospace MRO.

Capability	What Aerospace MRO Requires	What Traditional Planning Systems Deliver
Demand signal type	Forward-deterministic: maintenance events, AD compliance deadlines, flight schedules, and contract SLAs create knowable future demand before it occurs	Consumption-history model: statistical pattern detection on historical actuals, designed to extrapolate past behavior, not reason about future scheduled events
Inventory unit	Serialized unit: each part carries a serial number, maintenance history, airworthiness certification status, and documentation chain required for installation	Fungible SKU: aggregate quantities with no individual unit tracking, no certification status, no airworthiness documentation requirement
Stockout consequence	A stockout grounds an aircraft (AOG -- Aircraft on Ground) at $150K+ per hour. Holding an extra unit is a carrying cost. Running out is a crisis. These are not equivalent outcomes.	A stockout is a delayed order. Expedite routes and spot markets exist as a fallback. A miss is recoverable, just more expensive.
Supplier structure	Single-source with 12-18 month lead times, no spot market, potential export licensing requirements, when supply is exhausted, no alternative procurement path exists	Multi-source assumption: spot procurement and expedite are treated as available recovery options; lead time variability is a cost, not an absolute constraint
Org topology	Split demand and supply ownership: the airline knows the demand (maintenance program, tail utilization) and the MRO provider owns the supply obligation, demand signals must cross an organizational boundary	Single-org assumption: demand planner and supply planner share data within one ERP instance; no designed-in support for cross-organizational demand signal handoff

Why Now?

Three converging forces make solving this problem urgent and viable today.

01 Post-2020 supply chain fragility

The 2020 disruption exposed the fragility of lean MRO inventory strategies. Lead times that were measured in weeks stretched to 18 months. The industry spent two years in reactive mode and many operators are still rebuilding buffer stock. There is now genuine urgency around planning visibility that simply did not exist before 2020.

18 months

lead times at peak disruption

02 2025 tariff volatility

Components sourced from international suppliers face cost uncertainty that makes procurement planning harder. Demand accuracy, knowing what you will actually need before you have to order it, has become more valuable than it has ever been. Getting the forecast wrong now carries a higher dollar cost per miss.

$150K+/hr

cost of grounding one aircraft

03 700,000 technician shortage by 2043

The industry will not come close to filling the projected technician gap. Every hour a technician spends waiting for a part that should have been on the shelf is an hour of productive capacity permanently lost. Demand planning accuracy is now a workforce productivity problem, not just a logistics problem.

700,000

additional technicians needed by 2043

Who We Build For

We focus on the demand planning loop, where fragmented signals, cross-organizational handoffs, and manual reconciliation create the highest operational risk.

Persona	Current Daily Reality	What Changes with Partcast
Demand Analyst	Spends 60-70% of the planning cycle manually reconciling flight schedules, AD portals, ERP exports, and spreadsheets, producing a forecast that is stale before it reaches supply.	Arrives at the demand review with a draft forecast pre-generated, top signals already surfaced, and changes since last cycle flagged with explanations. The job shifts from data assembly to signal evaluation.
Supply Planner	Receives a consensus demand plan that is already stale; spends significant time in reactive mode expediting after AOG events occur rather than positioning inventory in advance of them.	Sees live AOG risk scores by tail number, immediately knows when an approved demand update changes their inventory position, and gets ranked repositioning recommendations with downstream trade-off analysis.
S&OP Leader	Spends the first hour of every S&OP meeting reconciling data, figuring out what the current plan actually is and why it changed, instead of making forward-looking allocation decisions.	Receives an AI-generated S&OP package 48 hours before the meeting: FVA history, root-cause attribution, AOG risk summary, and forward plan. The meeting becomes a decision forum, not a data reconciliation session.
VP MRO Operations	Learns about AOG risk when an aircraft is already grounded, not 30-60 days before. The leading indicators, a tail approaching C-check with a critical part not yet on order, are buried in systems no one is synthesizing.	Has a live view of tails ranked by AOG risk with the specific demand and inventory factors driving each score. Escalations arrive before events, not after. Forward planning position is visible without depending on the planning team to surface it.

Agentic layer supporting the above workflows

AD Compliance Demand Agent

Autonomous: Parses new ADs, identifies in-scope tails, calculates time-to-compliance and demand gap, and queues a recommendation to the planner.

Handoff: Hands off when the demand gap requires a procurement decision -- planner sees the full reasoning chain, per-tail calculations, and confidence flags before approving.

AOG Risk Scoring Agent

Autonomous: Re-scores AOG risk for every tail on a 6-hour cadence and immediately on unscheduled removal events, supplier lead time changes, or flight schedule shifts.

Handoff: Hands off at Alert (66+) with a demand recommendation; at Critical (86+) escalates to the S&OP Leader with a procurement countdown and time-sensitive action bar.

S&OP Package Agent

Autonomous: Computes cycle variances, classifies root causes (model error vs override error vs external shock), calculates FVA metrics, and assembles the full S&OP package.

Handoff: Hands off a live annotatable document, not a PDF, where the S&OP Leader records decisions on each flagged item and closes the cycle.

Product Strategy

The core strategy is to transform demand planning from a manual, reconciliation-driven process into an event-driven system that produces actionable, explainable recommendations.

MRO demand is largely knowable in advance, but fragmented across systems. The product prioritizes signal integration and decision workflows over raw data aggregation, enabling planners to focus on evaluating decisions rather than assembling inputs.

Strategic Principles

1. **Event-driven planning over historical forecasting**

   Demand is modeled based on maintenance events, AD compliance, and utilization -- not consumption history.

2. **Decision support over data aggregation**

   The system surfaces prioritized, actionable recommendations rather than exposing raw data.

3. **Explainability over black-box accuracy**

   Recommendations are backed by clear signal-level reasoning that planners can validate.

4. **AOG risk minimization as the primary objective**

   All decisions are framed around avoiding aircraft-on-ground events, not generic service levels.

5. **Human-in-the-loop learning**

   Planner overrides are captured as structured inputs to continuously improve the system.

Product Strategy

Strategic Trade Offs

Deterministic logic vs **machine learning**

The product starts with deterministic, event-driven modeling to ensure transparency and immediate value, with ML layered in as override and actuals data becomes available.

Actionability vs **completeness of data**

Recommendations are generated with available signals and confidence flags rather than delayed for perfect data completeness.

Guidance vs **full automation**

The system prioritizes guided decision-making, with planners approving or overriding recommendations rather than fully autonomous execution.

Depth vs **breadth of coverage**

Initial focus is on solving demand planning deeply for MRO operators, before expanding across additional workflows or verticals.

How We Measure Success

Success is defined by shifting from reactive, spreadsheet-driven planning to proactive risk management with measurable improvements in forecast accuracy, cycle time, and decision accountability.

Metric	Definition	Baseline Today	Target at 12 Months
Forecast Value Add (FVA)	% of cases where the AI-generated forecast is more accurate than the planner-overridden forecast, measured at actuals close	Unmeasured -- no attribution system exists	**AI outperforms human override in >15% more cases**
Aircraft-on-Ground (AOG) Risk Coverage	% of AOG events that were preceded by a Partcast early-warning signal within a 30-day window	0% -- no proactive risk scoring exists	**>80% of AOG events flagged in advance**
Demand Review Cycle Time	Calendar days from signal update to approved consensus demand plan	~10 days -- manual reconciliation across systems	**<2 days -- AI pre-builds recommendations**
Override Attribution Rate	% of planner overrides that include a structured rationale (reason category + explanation)	<10% -- overrides are largely undocumented	**>90% -- structured rationale captured by default**
Planner Adoption Rate	% of demand analysts using AI-generated forecasts as their starting point	0% -- tool does not exist	**>80% within 6 months of deployment**

How Partcast Works

Partcast is structured as a four-layer system that transforms fragmented demand signals into actionable, auditable planning decisions. The architecture handles both structured signals (e.g., flight schedules, maintenance events) and semi-structured signals (e.g., AD documents, supplier updates), while maintaining explainability and planner control.

Signal Ingestion Layer (1)

Ingests and normalizes demand signals from multiple sources: flight schedules and maintenance event calendars, airworthiness directives (FAA / EASA), supplier lead time data and contract SLAs, ERP historical consumption

Structured signals are processed deterministically. Semi-structured signals are parsed with AI and assigned confidence scores, with low-confidence data routed for planner review.

Demand Intelligence Engine (2)

Generates demand forecasts at the part number x tail number level using an event-driven model: reasons forward from maintenance schedules and compliance deadlines, constructs demand timelines and confidence ranges, detects anomalies between expected and actual consumption

AI is used for extracting structured data from AD documents and identifying deviations. Core demand logic remains deterministic and auditable.

Collaborative Planning Layer (UX) (3)

Presents AI-generated recommendations to planners through a structured workflow: prioritized demand review queue, signal-level explanations for each recommendation, approve / override / escalate actions with rationale capture

All decisions are logged, enabling forecast-to-actual attribution and continuous learning.

Demand-Supply Matching Engine (4)

Maps approved demand plans to serialized inventory positions: identifies supply gaps and scores AOG (Aircraft-on-Ground) risk per tail, generates repositioning and procurement recommendations, surfaces trade-offs across competing inventory needs

Outputs are decision-ready and tied directly to operational risk.

Where AI is Used (and Where It is Not)

Where AI is Used

1 Signal extraction from semi-structured data

AI is used to parse and structure inputs that are not natively machine-readable: airworthiness directives (ADs), supplier updates and lead time changes, contract SLA documents. This converts unstructured inputs into structured signals that can be used in demand planning.

2 Anomaly detection and pattern recognition

ML models monitor actual consumption vs event-driven forecasts to: detect unexpected removals or failure patterns, identify deviations from expected demand, surface signals that require planner review.

3 Continuous learning from planner behavior

AI learns from planner overrides and forecast vs actual outcomes. This improves forecast accuracy over time and adapts to operator-specific patterns.

Where AI is Not Used

Core demand modeling and planning logic remain deterministic:

- Demand derived from maintenance events and compliance timelines
- Time-to-compliance calculations
- Demand gap and coverage calculations
- AOG risk scoring logic

These components are rules-based and fully auditable.

Core Agents

Partcast uses a set of specialized agents to translate AI capabilities into actionable workflows. Each agent is responsible for a specific stage in the planning loop. Agents do not replace planners. They augment decision-making and automate data synthesis, while keeping the human in control.

AD Compliance Demand Agent

1

Transforms regulatory and operational inputs into structured demand signals

- Parses airworthiness directives (ADs) and maintenance requirements
- Identifies affected aircraft (tails) and required parts
- Generates demand recommendations with supporting context

Role in system:

Converts external rules and documents into actionable demand

AOG Risk Scoring Agent

2

Continuously monitors demand and supply conditions to surface operational risk

- Evaluates inventory coverage against upcoming demand events
- Identifies aircraft at risk of AOG based on timing and supply gaps
- Prioritizes high-risk items for planner review

Role in system:

Shifts planning from reactive to proactive risk management

S&OP Package Agent

3

Aggregates decisions and prepares structured outputs for planning cycles

- Compiles forecast changes, overrides, and key signals
- Summarizes drivers of change and associated risks
- Produces a complete S&OP-ready package for review

Role in system:

Replaces manual preparation with automated, explainable outputs

Design Principles

Deterministic first, AI-augmented where necessary

1

Core demand logic is based on structured, event-driven calculations. AI is applied selectively where it adds value (e.g., signal extraction, anomaly detection), ensuring early reliability and long-term extensibility.

Explainable outputs with signal-level reasoning

2

Every recommendation is supported by a transparent chain of signals (e.g., tail-level demand drivers, compliance deadlines, lead times). Planners evaluate reasoning, not opaque model outputs.

Human-in-the-loop decision making with full auditability

3

Planners approve, override, or escalate all recommendations. Each decision is logged with structured rationale, enabling traceability and accountability across the planning process.

Continuous learning from overrides and actual outcomes

4

Planner feedback and forecast-to-actual outcomes are captured as training signals. The system improves over time by learning from real-world decisions and their results.

Roadmap & Prioritization

Prioritization follows a phased approach grounded in two core dimensions: immediate operational value for planning teams and long-term platform leverage through data accumulation and learning loops.

	Phase 1	Phase 2	Phase 3	Phase 4
	Signal Foundation & Trust	AI Sensing & Learning	Supply Matching & Risk	Agentic Planning & Autonomy
Key Capabilities	Signal ingestion (ERP, schedules, ADs), event-driven demand baseline (part x tail), demand review UX, recommendation explanations, audit trail and override capture	ML models trained on overrides and actuals, anomaly detection, confidence ranges, FVA tracking, bias analytics	Demand-supply matching engine, serialized inventory mapping, AOG risk scoring per tail, repositioning recommendations, S&OP package generation	Autonomous agents for demand updates and risk monitoring, planner supervision controls, continuous planning loop, incremental autonomy based on performance
Value Delivered	**Immediate improvement in planning cycle time and visibility**	**Measurable improvement in forecast accuracy**	**Shift from reactive execution to proactive risk management**	**Significant reduction in manual planning effort**
Moat Being Built	Structured demand signal + override rationale dataset begins forming	Customer-specific learning loop tied to real planning behavior	Workflow integration into S&OP and supply decision processes	Institutional knowledge captured in system + high switching cost

Roadmap & Prioritization

Approach

Given the complexity of MRO demand planning and the need to build trust in an AI-assisted system, it is neither practical nor desirable to deliver full autonomy from day one. Instead, prioritization follows a phased approach grounded in two core dimensions:

- **Immediate operational value for planning teams** (cycle time reduction, visibility, AOG risk awareness)
- **Long-term platform leverage through data accumulation and learning loops**

This approach ensures early, tangible value to drive adoption and trust, while building the structured data and workflow foundation required for advanced AI capabilities.

Phase	Focus	Key Capabilities
Phase 1 Signal Foundation & Trust	Establish demand visibility and replace manual reconciliation workflows	Signal ingestion (ERP, schedules, ADs), event-driven demand baseline (part × tail), demand review UX, recommendation explanations, audit trail and override capture
Phase 2 AI Sensing & Learning	Improve forecast accuracy and introduce measurable learning	ML models trained on overrides and actuals, anomaly detection, confidence ranges, FVA tracking, bias analytics
Phase 3 Supply Matching & Risk Management	Connect demand planning to inventory decisions and AOG risk	Demand-supply matching engine, serialized inventory mapping, AOG risk scoring per tail, repositioning recommendations, S&OP package generation
Phase 4 Agentic Planning & Autonomy	Transition to continuous, exception-based planning	Autonomous agents for demand updates and risk monitoring, planner supervision controls, continuous planning loop, incremental autonomy based on performance

Roadmap & Prioritization

01 — 0-6 months
Signal Foundation

Scope

- Ingest ERP, flight schedules, and maintenance event data
- Build deterministic, event-driven demand baseline (per part x tail)
- Launch demand review UX with AI-generated recommendations and audit trail

Value delivered:
Immediate improvement in planning cycle time and visibility

Moat:
Structured demand signal + override rationale dataset begins forming

02 — 6-12 months
AI Sensing

Scope

- Train ML models on override and actuals data
- Add confidence ranges and anomaly detection
- Launch FVA tracking and bias analytics

Value delivered:
Measurable improvement in forecast accuracy

Moat:
Customer-specific learning loop tied to real planning behavior

03 — 12-18 months
Supply Matching

Scope

- Launch demand-supply matching engine
- Introduce AOG risk scoring per tail
- Generate repositioning recommendations with trade-off visibility
- Auto-generate S&OP planning package

Value delivered:
Shift from reactive execution to proactive risk management

Moat:
Workflow integration into S&OP and supply decision processes

04 — 18-24 months
Agentic Planning

Scope

- Deploy autonomous agents for demand updates and risk monitoring
- Enable planner supervision and controlled autonomy
- Expand system to continuous planning vs periodic cycles

Value delivered:
Significant reduction in manual planning effort

Moat:
Institutional knowledge captured in system + high switching cost

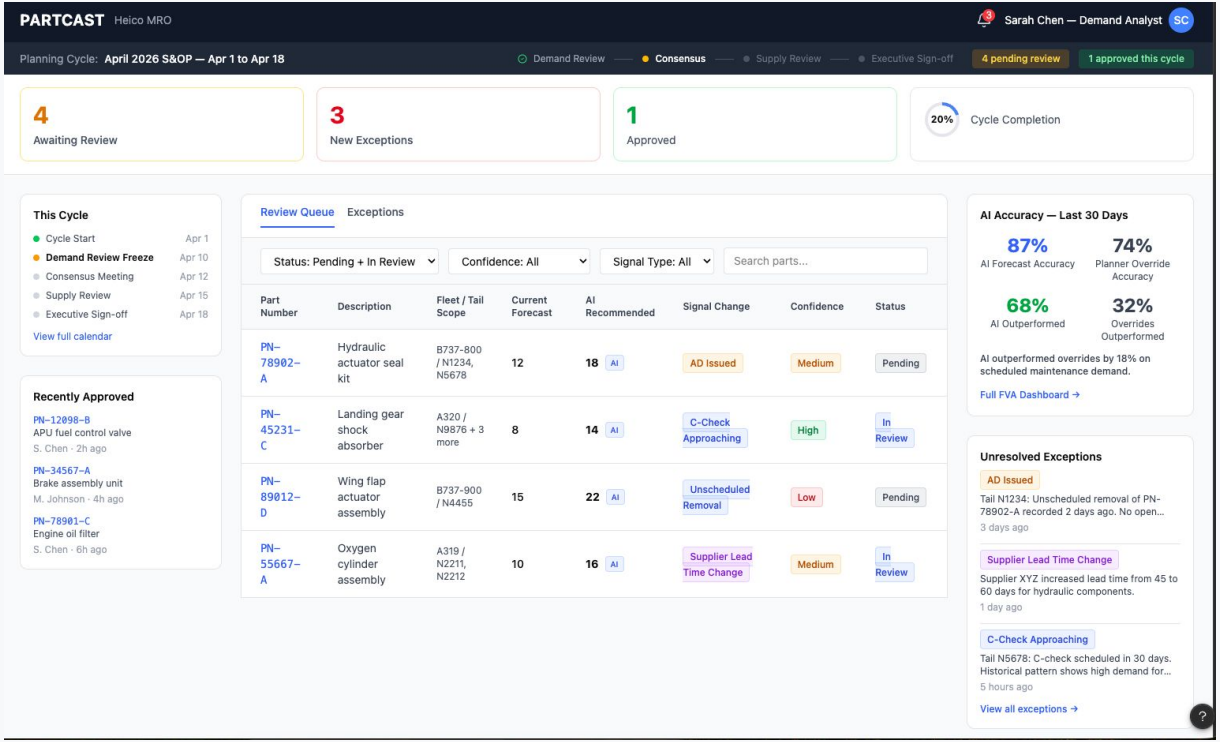
UX: Demand Review Queue

Purpose: The planner's daily starting point. Transforms fragmented demand signals into a prioritized queue of actionable decisions, replacing manual reconciliation.

Workflow

1. Assess cycle status -- view items awaiting review, new exceptions, and overall cycle progress
2. Work through prioritized queue -- each row represents a demand decision, ranked by urgency and signal change
3. Evaluate signals at a glance -- see key drivers (ADs, maintenance events, supplier updates) and affected tails
4. Focus using filters -- filter by status, confidence, or signal type to isolate high-priority items
5. Drill into decision -- select a row to evaluate recommendation and take action (approve / override / escalate)

Outcome: Reduces planning cycle time by shifting from manual data gathering to structured, AI-assisted decision-making



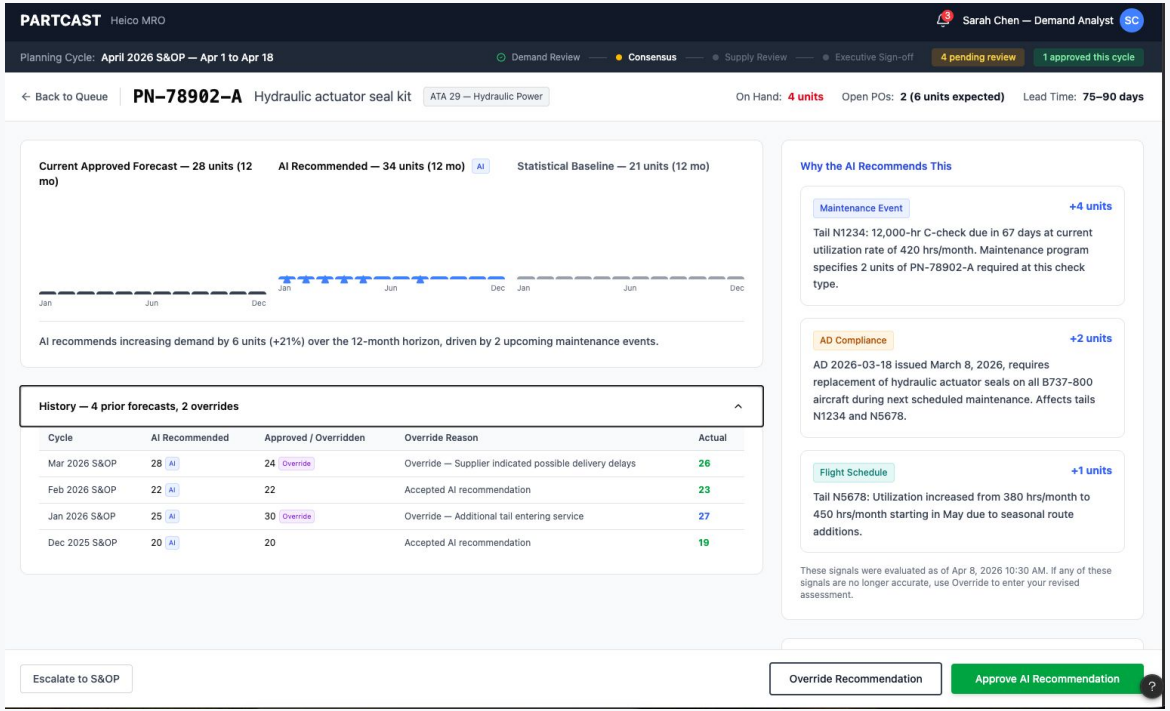
UX: Demand Recommendation Detail

Purpose: TThe core decision screen. Enables the planner to evaluate AI recommendations using signal-level reasoning and inventory risk scenarios, and take action (approve, override, or escalate).

Workflow

1. Understand forecast differences -- compare current plan, AI recommendation, and statistical baseline across time
2. Review signal evidence -- evaluate ranked signals (e.g., maintenance events, ADs, utilization) with per-signal demand impact
3. Assess inventory risk -- view stocking scenarios in operational terms (coverage, AOG exposure, carrying cost)
4. Reference historical decisions -- review past overrides and actual outcomes to inform current judgment
5. Take action -- approve AI recommendation, override with rationale, or escalate to S&OP

Outcome: Enables fast, explainable, and auditable decisions by shifting from intuition-based planning to evidence-driven, risk-aware decision-making



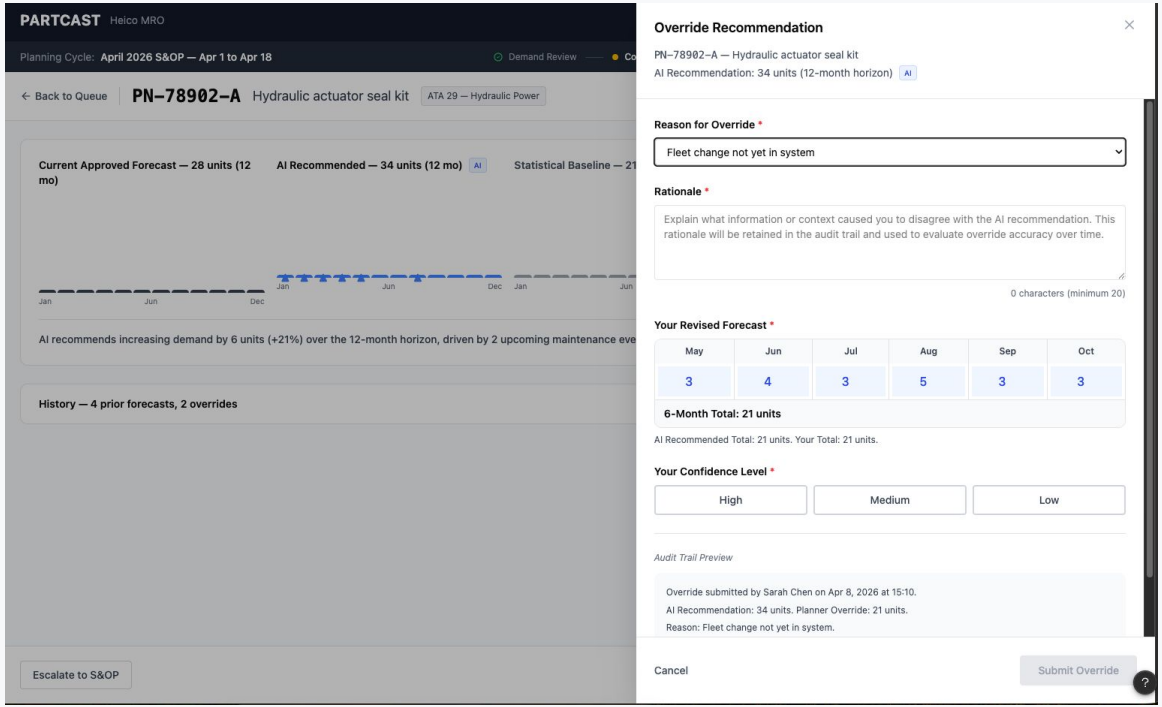
UX: Override Panel

Purpose: Enables planners to override AI recommendations with structured rationale and forecast adjustments, creating a complete audit trail and learning signal for the system.

Workflow

1. Select reason category -- classify override using standardized taxonomy (e.g., fleet change, supplier issue, AD update)
2. Provide detailed rationale -- capture context behind the decision in structured, auditable form
3. Adjust forecast values -- modify monthly forecast based on planner judgment and domain knowledge
4. Indicate confidence level -- record planner confidence to support future accuracy evaluation
5. Review audit trail preview and submit -- validate how the decision will be recorded before submitting

Outcome: Creates a structured dataset of planner decisions, rationale, and outcomes, enabling continuous learning, FVA measurement, and long-term model improvement



UX: FVA Dashboard

Purpose: Provides visibility into whether AI recommendations or planner overrides are more accurate over time, enabling data-driven decision-making and continuous improvement.

Workflow

1. Understand performance at a glance -- review headline insight summarizing AI vs planner accuracy over selected time period
2. Analyze accuracy trends over time -- compare AI, planner overrides, and statistical baseline across a rolling horizon
3. Identify strengths by demand type -- see where AI leads (e.g., scheduled maintenance) vs where planner judgment adds value
4. Review high-impact decisions - examine largest overrides and whether they improved or reduced forecast accuracy
5. Detect systematic bias -- identify over- or under-forecasting patterns across demand categories

Builds trust through transparent performance measurement, enabling planners to improve decisions over time and guiding when to rely on AI vs human judgment

Accuracy by Demand Event Type

Demand Event Type	AI Forecast	Planner Override	Verdict	# of Events
Scheduled Maintenance	92%	78%	AI leads	145
AD Compliance	88%	85%	AI leads	67
Unscheduled Removal	68%	84%	Override leads	89
Contract-Driven Demand	91%	90%	No difference	34
Aggregate	**87%**	**79%**	**AI leads**	**335**

Most Impactful Overrides

PN-78982-A — AI more accurate
Hydraulic actuator seal kit
AI: 34 units → Override: 28 units (-6 units)
Contract amendment pending

PN-45231-C — Override more accurate
Landing gear shock absorber
AI: 22 units → Override: 30 units (+8 units)
Fleet change not yet in system

PN-12898-B — AI more accurate
APU fuel control valve
AI: 18 units → Override: 12 units (-6 units)
Supplier reliability concern

PN-34521-F — AI more accurate
Engine bleed air valve
AI: 28 units → Override: 22 units (-6 units)
Supplier reliability concern

PN-67898-B — Override more accurate
Cabin pressure sensor
AI: 14 units → Override: 20 units (+6 units)
Fleet change not yet in system

PN-11223-C — AI more accurate
Fuel pump assembly
AI: 30 units → Override: 24 units (-6 units)
Contract amendment pending

PN-99887-A — AI more accurate
Avionics cooling fan
AI: 19 units → Override: 15 units (-4 units)
Supplier reliability concern

PN-44556-D — Override more accurate
Hydraulic reservoir
AI: 12 units → Override: 18 units (+6 units)
AD compliance timeline changed

In this period, 6 of the 10 most impactful overrides resulted in less accurate forecasts than the AI recommendation.

Forecast Bias

Aggregate	Scheduled Maintenance	AD Compliance	Unscheduled Removal	Contract-Driven
+2%	+3%	-2%	+15%	+2%
Within expected range	Within expected range	Within expected range	Over by 15%	Within expected range
Driven by AI	Driven by AI	Driven by Override	Driven by Override	Driven by AI

Bias measures whether forecasts are consistently too high (over-forecasting) or too low (under-forecasting) relative to actuals. Persistent bias in either direction suggests a systematic input signal issue.

Why Partcast Wins

Owns the planning workflow (not just data)

Partcast replaces the most critical and underserved workflow in MRO planning -- the demand review cycle. Instead of acting as a reporting layer, it becomes the system where planners evaluate, decide, and act, embedding directly into daily operations.

Builds a proprietary decision dataset

Every interaction generates structured data: demand signals (ADs, schedules, utilization), planner overrides and rationale, forecast vs actual outcomes. This creates a customer-specific dataset of decisions and outcomes that cannot be replicated without sustained usage.

Improves continuously through human-in-the-loop learning

The system learns from when planners override, when AI is correct, and where each approach performs better. Over time, this leads to higher forecast accuracy (FVA), better calibration of AI recommendations, and increasing planner trust and adoption.

Creates durable switching costs through workflow and trust

Once embedded: planning cycles depend on Partcast outputs, S&OP processes align around its recommendations, historical decisions and rationale are stored within the system. Switching away means losing accumulated learning, audit history, and decision context.

Expands from wedge to platform

Starting with demand review, Partcast expands into supply matching and AOG risk management, S&OP decision support, and continuous, agent-driven planning. This creates a path from single workflow to core planning platform.